Lie Theory: Chapter 3 Homework

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1. Let L be a non-abelian 3-dimensional Lie algebra with basis $\{f, g, z\}$ and $[f, g] \coloneqq z$. Assume $z \in Z(L)$ and L' is one-dimensional. Prove that $L' = Z(L)$.

Proof. First, we realize that a general element of L' is

$$a[f, g] + b[f, z] + c[g, z] = az$$

because $[f, g] = z$ and $[f, z] = [g, z] = 0$ since z is in the center. Since we are given that L' is one-dimensional, it must be spanned by z. Thus when we bracket a general element of L' with a general element of L:

$$[\alpha z, af + bg + cz] = \alpha[z, af + bg + cz] = 0$$

since $z \in Z(L)$. Thus $L' \subset Z(L)$.

Now we examine a general element of the center, $af + bg + cz$, bracketed with a general element of L:

$$\begin{aligned} 0 &= [af + bg + cz, \alpha f + \beta g + \gamma z] \\ &= a\beta[f, g] + a\gamma[f, z] + b\alpha[g, f] + b\gamma[g, z] + c\alpha[z, f] + c\beta[z, g] \\ &= a\beta z - b\alpha z \\ &= (a\beta - b\alpha)z. \end{aligned}$$

Since $z \neq 0$, we have $a\beta = b\alpha$ *for every choice of* α *and* β. In particular, if $\alpha = 0$ and $\beta = 1$, we get that $a = 0$, and if $\alpha = 1$ but $\beta = 0$, we have $0 = b$. Thus $af + bg + cz = cz$, where c is truly free to vary over F, so $Z(L)$ is one-dimensional. We already showed that $L' \subset Z(L)$, so since they have the same dimension, $L' = Z(L)$. □

2. Check that the strictly upper triangular matrices $\{e_{12}, e_{23}, e_{13}\}$ satisfy the following properties.

(a) $[e_{12}, e_{23}] = e_{13}$.

(b) e_{13} is in the center of the Lie algebra of strictly upper triangular matrices.

This shows that the set $\{e_{12}, e_{23}, e_{13}\}$ satisfies the construction described in Section 3.2.1 with $z = e_{13}$ and f and g equal to e_{12} and e_{23}.

Proof.

(a)

$$\begin{aligned}
[e_{12}, e_{23}] &= \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \\
&= \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \\
&= \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \\
&= e_{13}.
\end{aligned}$$

(b) We bracket e_{13} with a general element of $b(3, F)$:

$$\begin{aligned}
\left[\begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & a & b \\ 0 & 0 & c \\ 0 & 0 & 0 \end{pmatrix} \right] &= \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & a & b \\ 0 & 0 & c \\ 0 & 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & a & b \\ 0 & 0 & c \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \\
&= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \\
&= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}
\end{aligned}$$

so $e_{13} \in Z(n(3, F))$ as desired.

□

3. Let V be a vector space and let φ be an endomorphism of V. Let L have underlying vector space $V \oplus \mathrm{Span}\{x\}$. Show that if we define the Lie bracket on L by $[y, z] = 0$ and $[x, y] = \varphi(y)$ for $y, z \in V$, then L is a Lie algebra and $\dim(L') = \mathrm{rank}(\varphi)$.

Proof. Given any $y, z \in V$ and $a, b \in F$, we represent a general bracket in L as follows:

$$\begin{aligned}
[(y, ax), (z, bx)] &= [(y, 0) + (0, ax), (z, 0) + (0, bx)] \\
&= [(y, 0), (z, 0)] + [(y, 0), (0, bx)] + [(0, ax), (z, 0)] + [(0, ax), (0, bx)] \\
&= 0 + [(y, 0), b(0, x)] + [a(0, x), (z, 0)] + [a(0, x), b(0, x)] \\
&= b[(y, 0), (0, x)] + a[(0, x), (z, 0)] + ab[(0, x), (0, x)] \\
&= -b\varphi(y) + a\varphi(z) \\
&= a\varphi(z) - b\varphi(y) \\
&= (a\varphi(z) - b\varphi(y), 0) \\
&= (\varphi(az - by), 0),
\end{aligned}$$

where we will frequently use the abuse-of-notation third-to-last line for brevity. Now we can easily check the conditions for a bracket (bilinearity is already assumed). First,

$$[(y, ax), (y, ax)] = a\varphi(y) - a\varphi(y) = 0$$

as desired, and

$$\begin{aligned}
&[(w, ax), [(y, bx), (z, cx)]] + [(y, bx), [(z, cx), (w, ax)]] + [(z, cx), [(w, ax), (y, bx)]] \\
&= [(w, ax), b\varphi(z) - c\varphi(y)] + [(y, bx), c\varphi(w) - a\varphi(z)] + [(z, cx), a\varphi(y) - b\varphi(w)] \\
&= [(w, ax), (b\varphi(z) - c\varphi(y), 0)] + [(y, bx), (c\varphi(w) - a\varphi(z), 0)] + [(z, cx), (a\varphi(y) - b\varphi(w), 0)] \\
&= a\varphi(b\varphi(z) - c\varphi(y)) + b\varphi(c\varphi(w) - a\varphi(z)) + c\varphi(a\varphi(y) - b\varphi(w)) \\
&= ab\varphi^2(z) - ac\varphi^2(y) + bc\varphi^2(w) - ab\varphi^2(z) + ac\varphi^2(y) - bc\varphi^2(w) \\
&= 0,
\end{aligned}$$

where the final equality comes from canceling like terms. Thus L is a Lie algebra.

Returning to our general formulation of the bracket in L where

$$[(y, ax), (z, bx)] = (\varphi(az - by), 0),$$

we see that if $z \in V$, then $[(0, x), (z, 0)] = (\varphi(z), 0)$, so $\operatorname{im}\varphi \times \{0\} \subset L'$, and we have $\dim(\operatorname{im}\varphi \times \{0\}) = \dim \operatorname{im}\varphi = \operatorname{rank}(\varphi) \leq \dim(L')$. On the other hand, we have shown that the brackets of L have the form $(\varphi(v), 0)$ for some $v \in V$, so any element of L' is a linear combination of such elements, namely something like this:

$$(\varphi(v_1 + \cdots + v_n), 0).$$

Thus $L' \subset \operatorname{im}\varphi \times \{0\}$,

$$\dim(L') \leq \dim(\operatorname{im}\varphi \times \{0\}) = \dim \operatorname{im}\varphi = \operatorname{rank}\varphi$$

Putting the two inequalities together, we get that $\dim(L') = \operatorname{rank}(\varphi)$ as desired. □

4. Suppose L is a vector space with basis $\{x, y\}$ and that a bilinear operation on L is defined such that $[u, u] = 0$ for all $u \in L$. Show that the Jacobi identity holds and hence L is a Lie algebra.

Proof. Since $[u, u] = 0$ for all $u \in L$, we have the familiar fact that $[u, v] = -[v, u]$. We also still have the core fact about structure constants, namely that the proposed bracket depends entirely on $[x, y]$. Let α, β be the constants satisfying $[x, y] = \alpha x + \beta y$. Thus we can go ahead and compute that $[x, [x, y]] = [x, \alpha x + \beta y] = \beta[x, y]$, and similarly $[y, [x, y]] = [y, \alpha x + \beta y] =$

$-\alpha[x,y]$. We have

$$\begin{aligned}
&[a_1x+b_1y,[a_2x+b_2y,a_3x+b_3y]]\\
&\quad+[a_2x+b_2y,[a_3x+b_3y,a_1x+b_1y]]\\
&\quad+[a_3x+b_3y,[a_1x+b_1y,a_2x+b_2y]]\\
&=[a_1x+b_1y,(a_2b_3-a_3b_2)[x,y]]\\
&\quad+[a_2x+b_2y,(a_3b_1-a_1b_3)[x,y]]\\
&\quad+[a_3x+b_3y,(a_1b_2-a_2b_1)[x,y]]\\
&=[a_1x,(a_2b_3-a_3b_2)[x,y]]+[b_1y,(a_2b_3-a_3b_2)[x,y]]\\
&\quad+[a_2x,(a_3b_1-a_1b_3)[x,y]]+[b_2y,(a_3b_1-a_1b_3)[x,y]]\\
&\quad+[a_3x,(a_1b_2-a_2b_1)[x,y]]+[b_3y,(a_1b_2-a_2b_1)[x,y]]\\
&=a_1(a_2b_3-a_3b_2)[x,[x,y]]+b_1(a_2b_3-a_3b_2)[y,[x,y]]\\
&\quad+a_2(a_3b_1-a_1b_3)[x,[x,y]]+b_2(a_3b_1-a_1b_3)[y,[x,y]]\\
&\quad+a_3(a_1b_2-a_2b_1)[x,[x,y]]+b_3(a_1b_2-a_2b_1)[y,[x,y]]\\
&=\alpha a_1(a_2b_3-a_3b_2)[x,y]-\beta b_1(a_2b_3-a_3b_2)[x,y]\\
&\quad+\alpha a_2(a_3b_1-a_1b_3)[x,y]-\beta b_2(a_3b_1-a_1b_3)[x,y]\\
&\quad+\alpha a_3(a_1b_2-a_2b_1)[x,y]-\beta b_3(a_1b_2-a_2b_1)[x,y]\\
&=(\alpha a_1a_2b_3-\alpha a_1a_3b_2-\beta b_1a_2b_3+\beta b_1a_3b_2\\
&\quad+\alpha a_2a_3b_1-\alpha a_2a_1b_3-\beta b_2a_3b_1+\beta b_2a_1b_3\\
&\quad+\alpha a_3a_1b_2-\alpha a_3a_2b_1-\beta b_3a_1b_2+\beta b_3a_2b_1)[x,y]\\
&=0[x,y]\\
&=0,
\end{aligned}$$

where second-to-last equality comes from canceling out like terms. Thus the Jacobi identity is satisfied. □

5. Suppose that I is an ideal of a Lie algebra L and that there is a subalgebra S of L such that $L=S\oplus I$. Show that the map $\theta:S\to\mathfrak{gl}(I)$ defined by $\theta(s)(x)=[s,x]$ is a Lie algebra homomorphism from S into $\mathrm{Der}(I)$.

We say that L is a *semidirect product* of I by S.

Proof. First we show that $\theta(s)$ is a derivation on I for all $s\in S$ by first noting that indeed $[s,x]\in I$ whenever $x\in I$ since I is an ideal, then showing that it is linear by

$$\begin{aligned}
\theta(s)(ax+by)&=[s,ax+by]\\
&=a[s,x]+b[s,y]\\
&=a\theta(s)(x)+b\theta(s)(y)
\end{aligned}$$

as desired, then showing that it satisfies the Leibniz rule:

$$\begin{aligned}
\theta(s)([x,y])&=[s,[x,y]]\\
&=[x,[s,y]]+[[s,x],y]\\
&=[x,\theta(s)(y)]+[\theta(s)(x),y].
\end{aligned}$$

Now we show that θ is linear:

$$\begin{aligned}\theta(as_1+bs_2)(x) &= [as_1+bs_2, x] \\ &= a[s_1,x]+b[s_2,x] \\ &= a\theta(s_1)(x)+b\theta(s_2)(x),\end{aligned}$$

and finally we show that θ preserves the bracket:

$$\begin{aligned}\theta([s_1,s_2])(x) &= [[s_1,s_2],x] \\ &= -[x,[s_1,s_2]] \\ &= [s_1,[s_2,x]]+[s_2,[x,s_1]] \\ &= [s_1,[s_2,x]]-[s_2,[s_1,x]] \\ &= \theta(s_1)([s_2,x])-\theta(s_2)([s_1,x]) \\ &= \theta(s_1)(\theta(s_2)(x))-\theta(s_2)(\theta(s_1)(x)) \\ &= (\theta(s_1)\circ\theta(s_2))(x)-(\theta(s_2)\circ\theta(s_1))(x) \\ &= (\theta(s_1)\circ\theta(s_2)-\theta(s_2)\circ\theta(s_1))(x) \\ &= [\theta(s_1),\theta(s_2)](x).\end{aligned}$$

Thus θ is a Lie algebra homomorphism $S \to \mathrm{Der}(I)$ as desired. □

6. Show conversely that given Lie algebras S and I and a Lie algebra homomorphism $\theta : S \to \mathrm{Der}(I)$, the vector space $S \oplus I$ may be made into a Lie algebra by defining

$$[(s_1,x_1),(s_2,x_2)] = ([s_1,s_2],[x_1,x_2]+\theta(s_1)x_2-\theta(s_2)x_1)$$

for $s_1, s_2 \in S$, and $x_1, x_2 \in I$, and that this Lie algebra is a semidirect product of I by S. (The direct sum construction introduced in Exercise 2.6 is the special case where $\theta(s) = 0$ for all $s \in S$.)

Proof. First we check that the proposed bracket is bilinear:

$$\begin{aligned}&[a(s_1,x_1)+b(s_2,x_2),(s_3,x_3)] \\ &= [(as_1+bs_2,ax_1+bx_2),(s_3,x_3)] \\ &= ([as_1+bs_2,s_3],[ax_1+bx_2,x_3]+\theta(as_1+bs_2)x_3-b\theta(s_3)x_2) \\ &= (a[s_1,s_3]+b[s_2,s_3],a[x_1,x_3]+b[x_2,x_3]+a\theta(s_1)x_3+b\theta(s_2)x_3-b\theta(s_3)x_2) \\ &= (a[s_1,s_3],a[x_1,x_3]+a\theta(s_1)x_3-a\theta(s_3)x_1)+(b[s_2,s_3],b[x_2,x_3]+b\theta(s_2)x_3-b\theta(s_3)x_2) \\ &= a([s_1,s_3],[x_1,x_3]+\theta(s_1)x_3-\theta(s_3)x_1)+b([s_2,s_3],[x_2,x_3]+\theta(s_2)x_3-\theta(s_3)x_2) \\ &= a[(s_1,x_1),(s_3,x_3)]+b[(s_2,x_2),(s_3,x_3)].\end{aligned}$$

The proof follows similarly for the second slot.

Now we check the two bracket conditions. First,

$$\begin{aligned}[(s,x),(s,x)] &= ([s,s],[x,x]+\theta(s)x-\theta(s)x) \\ &= ([s,s],[x,x]) \\ &= (0,0).\end{aligned}$$

Now to check the Jacobi identity, which we write as $J((s_1,x_1),(s_2,x_2),(s_3,x_3)) = 0$ for simplicity, we note that J is linear in each component, so we can break this up into

$$\begin{aligned}0 =& J((s_1,0),(s_2,0),(s_3,0)) + J((0,x_1),(0,x_2),(0,x_3))\\ &+J((s_1,0),(s_2,0),(0,x_3)) + J((0,x_1),(0,x_2),(s_3,0))\\ &+J((s_1,0),(0,x_2),(0,x_3)) + J((0,x_1),(s_2,0),(s_3,0))\\ &+J((s_1,0),(0,x_2),(s_3,0)) + J((0,x_1),(s_2,0),(0,x_3)).\end{aligned}$$

We also note that the Jacobi identity is cyclically symmetric, i.e.

$$J(x,y,z) = J(z,x,y) = J(y,z,x),$$

so we need only check terms that are distinct with respect to cyclic permutation, namely

$$\begin{aligned}&J((s_1,0),(s_2,0),(s_3,0)),\\ &J((0,x_1),(0,x_2),(0,x_3)),\\ &J((s_1,0),(s_2,0),(0,x_3)),\\ &J((0,x_1),(0,x_2),(s_3,0)).\end{aligned}$$

If each of these is zero individually, the eight-term decomposition of J is zero. The first two are immediate since $J((s_1,0),(s_2,0),(s_3,0))$ reduces to the Jacobi identity in S, and likewise $J((0,x_1),(0,x_2),(0,x_3))$ reduces to the Jacobi identity in I. For the last two, then, we will use the following general rules:

$$\begin{aligned}[(s_1,0),(s_2,0)] &= ([s_1,s_2],[0,0] + \theta(s_1)0 - \theta(s_2)0)\\ &= ([s_1,s_2],0+0-0)\\ &= ([s_1,s_2],0),\\ [(0,x_1),(0,x_2)] &= ([0,0],[x_1,x_2] + \theta(0)x_2 - \theta(0)x_1)\\ &= (0,[x_1,x_2]+0-0)\\ &= (0,[x_1,x_2]),\\ [(s,0),(0,x)] &= ([s,0],[0,x] + \theta(s)x - \theta(0)0)\\ &= (0,0+\theta(s)x-0)\\ &= (0,\theta(s)x), \text{ and}\\ [(0,x),(s,0)] &= -[(s,0),(0,x)]\\ &= (0,-\theta(s)x).\end{aligned}$$

We have

$$\begin{aligned}
J((s_1,0),(s_2,0),(0,x_3)) &= [(s_1,0),[(s_2,0),(0,x_3)]] \\
&\quad + [(s_2,0),[(0,x_3),(s_1,0)]] \\
&\quad + [(0,x_3),[(s_1,0),(s_2,0)]] \\
&= [(s_1,0),(0,\theta(s_2)x_3)] \\
&\quad + [(s_2,0),(0,-\theta(s_1)x_3)] \\
&\quad + [(0,x_3),([s_1,s_2],0)] \\
&= (0,\theta(s_1)(\theta(s_2)x_3)) \\
&\quad + (0,\theta(s_2)(-\theta(s_1)x_3)) \\
&\quad + (0,-\theta([s_1,s_2])x_3) \\
&= (0,\theta(s_1)\theta(s_2)x_3 - \theta(s_2)\theta(s_1)x_3 - [\theta(s_1),\theta(s_2)]x_3) \\
&= (0,\theta(s_1)\theta(s_2)x_3 - \theta(s_2)\theta(s_1)x_3 - (\theta(s_1)\theta(s_2) - \theta(s_2)\theta(s_1))x_3) \\
&= (0,\theta(s_1)\theta(s_2)x_3 - \theta(s_2)\theta(s_1)x_3 - \theta(s_1)\theta(s_2)x_3 + \theta(s_2)\theta(s_1)x_3) \\
&= (0,0),
\end{aligned}$$

and

$$\begin{aligned}
J((0,x_1),(0,x_2),(s_3,0)) &= [(0,x_1),[(0,x_2),(s_3,0)]] \\
&\quad + [(0,x_2),[(s_3,0),(0,x_1)]] \\
&\quad + [(s_3,0),[(0,x_1),(0,x_2)]] \\
&= [(0,x_1),(0,-\theta(s_3)x_2)] \\
&\quad + [(0,x_2),(0,\theta(s_3)x_1)] \\
&\quad + [(s_3,0),(0,[x_1,x_2])] \\
&= (0,[x_1,-\theta(s_3)x_2]) \\
&\quad + (0,[x_2,\theta(s_3)x_1]) \\
&\quad + (0,\theta(s_3)[x_1,x_2]) \\
&= (0,-[x_1,\theta(s_3)x_2]) \\
&\quad + (0,[x_2,\theta(s_3)x_1]) \\
&\quad + (0,[\theta(s_3)x_1,x_2] + [x_1,\theta(s_3)x_2]) \\
&= (0,-[x_1,\theta(s_3)x_2] + [x_2,\theta(s_3)x_1] + [\theta(s_3)x_1,x_2] + [x_1,\theta(s_3)x_2]) \\
&= (0,-[x_1,\theta(s_3)x_2] + [x_2,\theta(s_3)x_1] - [x_2,\theta(s_3)x_1] + [x_1,\theta(s_3)x_2]) \\
&= (0,0).
\end{aligned}$$

Thus all the terms in the expansion of $J((s_1,x_1),(s_2,x_2),(s_3,x_3))$ are zero, so the Jacobi identity holds. □